

MAURICIO BARBA DA COSTA

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EDUCATION

Massachusetts Institute of Technology

MEng: AI & Decision-making

BS: Mathematics (18) and AI & Decision-making (6-4)

- **MEng GPA: 5.0/5.0** **BS GPA: 4.9/5.0**

Cambridge, MA

September 2024 – January 2026

September 2020 – May 2024

SELECT PUBLICATIONS

Faster LLM Inference via Sequential Monte Carlo

Yahya Emara, Mauricio Barba da Costa*, et al.*

90th percentile of reviews at COLM, 2026.

- Conceived of the novel speculative decoding algorithm and led a team to build a high-performance inference engine that is **5.2x** faster than autoregressive decoding.

Improving Autoformalization via Cycle Consistency and Incremental Type-Checking Using Language-Model Probabilistic Programs

Mauricio Barba da Costa, Fabian Zaiser*, et al.*

NeurIPS MATH-AI Workshop, 2025.

- Accelerated decode speed of the [GenLM](#) constrained generation inference engine, achieving **2.5x** lower latency and 22% higher accuracy than autoregressive decoding in the formal verification domain.

People Use Fast, Flat Goal-Directed Simulation to Reason about Novel Problems

Katherine M. Collins, Cedegao E. Zhang*, Lionel Wong*, Mauricio Barba da Costa*, Graham Todd*, et al.*

Accepted to Nature, 2026.

- Accelerated heuristic search algorithm by **20x** by leveraging GPU compute with JAX, conceived of hypotheses about human cognition, and led statistical analyses.

WORK EXPERIENCE

GenWeb

Research Engineer

Cambridge, MA

February 2026 – Current

- Building high-performance, low-latency systems at the intersection of language models, databases, and probabilistic inference at stealth startup founded by Reid Hoffman and Vikash Mansinghka.
- Work under NDA.

MIT 6.390 (IntroML)

Head TA

New York City, NY

September 2024 – December 2025

- Led a team of 80 TAs to instruct a class of 400+ students. Tasks included devising new content goals, updating the course website, and teaching recitations.
- Managed a team of 20 TAs to update our website with new course material. (<https://introml.mit.edu>).
- Reduced average response time on question-answering platform from 2 hours to 23 minutes and reduced turnaround time to fix bugs on website from weeks to days.

Goldman Sachs

Quantitative Research Intern

New York City, NY

June 2023 – August 2023

- Developed two low-latency signals for equities trading based on vector embeddings and transformer time series prediction.
- Leveraged multi-GPU clusters for training models with PyTorch.
- Return offer extended.

Fabri

Co-founder and CTO

Cambridge, MA

August 2021 – June 2023

- Led team of four to build Fabri, a startup that 3d prints wax patterns for the metal casting industry.
- Raised \$25,000 in funding from MIT's Sandbox innovation fund and \$200,000 from an angel investor.
- Established partnerships with foundries to cast our 3d prints parts and secured 2 contracts worth \$20k.
- Built system firmware for 3d printer and developed website leading to successful pitch demos to investors, the Sandbox program, and the NSF Spark i-Corps program.

Blockdaemon

Research Intern

Miami, FL

May 2022 – August 2022

- Engineered and cleaned data from Remote Procedure Call (RPC) nodes.
- Used generalized linear models to predict future returns of staking protocols and detect trends in Ethereum mining, leading to increased in prediction accuracy by 20%.

EXTRACURRICULAR INTERESTS

- Long-distance biking – Biked from San Francisco to LA and from Barcelona to Florence.
- Music – Play piano for friends and family often and raised \$5,000 to replace MIT Theta Chi piano.